



CONTACT

LOCATION
Bandung, West Java

WHATSAPP
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GMAIL
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LINKEDIN
linkedin.com/in/aliyushedri/

GITHUB
github.com/aliyushedri

PORTFOLIO
aliyushedri-portfolio.tech

SCOPE
Remote work; onsite by discussion.

LANGUAGES

Indonesian - Native

English - Intermediate / Working proficiency

SKILLS

Programming
C/C++, Python, MATLAB, JavaScript

Robotics & SLAM
ROS/ROS2, FAST-LIO, SLAM, sensor fusion, mobile & legged robots

Embedded & IoT
Arduino, ESP32, STM32, Raspberry Pi, MQTT, HTTP/REST

AI & automation
OpenCV, YOLOv8, ML/DL, TensorFlow/PyTorch, LLM and Telegram bots

Android & web
Android Studio, HTML/CSS/JS, GitHub Pages

Electronics & research
EasyEDA, EasyEDA Pro / PCB, Git/Linux, LaTeX, technical writing

ALIYUS HEDRI

Freelance Robotics, Embedded Systems & IoT

SUMMARY

Electrical Engineering Master by Research student at Telkom University, conducting research with ACS Laboratory. Focused on robotics, LiDAR-inertial odometry / SLAM, embedded systems, IoT, automation, and AI. Available for freelance and project-based work. Remote work; onsite by discussion.

EDUCATION

Master by Research in Electrical Engineering - Telkom University Ongoing
ACS Laboratory, Bandung.

Bachelor of Engineering in Electrical Engineering - Telkom University 2022 - 2026
GPA 3.8 / 4.0 - APERTI BUMN Scholarship Awardee 2022.

SMA Negeri Plus Provinsi Riau, Pekanbaru 2019 - 2022
Science major.

INTERNSHIP

Mechatronics Engineer - PT Arthatronic Studio Teknologi 30 Jun - 9 Aug 2025

SELECTED PROJECTS

TEL-U BOT Android application 2026
Published Android app built with Android Studio, supporting lecturer research through local Wi-Fi UDP control of an ESP32 robot, actuator commands, line/LiDAR/odometry telemetry, calibration, planner and JSON tools. [Google Play](#) - TEL-U BOT

Final-year project - 3D SLAM for AGV using FAST-LIO 2026
Adapted FAST-LIO for four Kinect V1 RGB-D cameras on ROS2 Jazzy with an STM32 + Mini PC distributed architecture.

TIRNARA - IoT fish-farming water-quality monitor 2025
Portable biofloc water-quality monitor using IoT and renewable energy; PIMNAS 2025 participant, PKM-KI.

Household water-quality monitoring system Feb - Jun 2025
ESP32 with pH, temperature, and turbidity sensors; real-time MQTT / HTTP cloud dashboard.

ARJUNA hexapod robot 2023 - 2024
Led development of a six-legged robot with IMU balance and real-time YOLOv8 detection; KRI 2024 SAR finalist.

TOMOT plant-monitoring stick (PKM-KI) Mar - Nov 2024
Portable pH, moisture, NPK, and temperature monitor; Bronze Medal, PIMNAS 37 presentation.

Bilingual portfolio website 2026
Built an HTML / CSS / JavaScript GitHub Pages portfolio with bilingual project pages, certificates, and downloadable CV. aliyushedri-portfolio.tech

ORGANIZATION

Lab Coordinator - EIRRG (Electronics & Intelligence Robotics Research Group) 2024 - 2025

Basic Computer Laboratory Assistant 2023 - 2025

Committee Member - Telkom University Student Orientation (PKKMB) 2023

AWARDS & ACHIEVEMENTS

- Top 20 Outstanding Students in Student Affairs - Telkom University, 2024.
- Finalist, Kontes Robot Indonesia 2024 SAR division - Team Leader, ARJUNA.
- Bronze Medal, PIMNAS 37 (2024) presentation - PKM-KI, TOMOT.
- Participant, PIMNAS 2025 - PKM-KI, TIRNARA.
- 2nd Place, Laboratory Competency Contest - Pekan Raya Biologi 2020, Universitas Riau.
- Participant, Madrasah Science Competition (KSM) 2021, Biology, Riau.